Do Large Language Models Have Style? A Stylometric Analysis of LLM Authorship

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Abstract

This study investigates whether different Large Language Models have an identifiable writing style of their own. In this experimental study, stylistics responses to dedicated prompts of 5 LLMs are compared, 3 of them closed-source models which are GPT-4o-mini, Claude Haiku 4.5, Gemini 2.5 Flash and 2 opensource models which are Qwen2.5-7B and LLaMA 3.1-8B. From 6 genres: explanation, opinion, comparison, argumentation, narration and dialogue and 120 prompts in total for 20 each, a corpus is created with 600 answers of LLMs. Quantitative results of answers of LLMs are created with using spaCy and Stylometrix, and to classify them, Random Forest model is trained. This classifier achieves 54% accuracy which are significantly higher than 20% random baseline. SHAP analysis shows that lexical diversity, average sentence length and punctuation frequency are the most discriminative stylistic features. t-SNE visualization also shows that, the genre of the text has impact on stylistic separability. These results demonstrate that LLMs have identifiable stylistic signatures, but how consistent these patterns are depends on the type of discourse they are generating.

Keywords: Large Language Models, Stylometry, Authorship Attribution, Natural Language Processing, Random Forest

1 Introduction

The rapid development of Large Language Models has raised an interesting question: do these models have their own writing style? Stylometry, the computational analysis of writing style, has been used for human authorship attribution for a long time. While previous work has focused on distinguishing human from AI-generated text [2, 3], less attention has been paid to whether different LLMs can be distinguished from each other. This project investigates this question by treating each LLM as a potential author with its own stylistic identity. The corpus is generated from five LLMs across six discourse genres, extract stylometric features using spaCy and StyloMetrix [1], and train a classifier to perform authorship attribution among models.

2 Research Question and Methodology

2.1 Research Question

The central research question of this project is: *Do Large Language Models exhibit distinctive and identifiable stylistic fingerprints that persist across different discourse genres?*

This question can be decomposed into the following sub-questions:

- Can a classifier trained on stylometric features reliably attribute texts to their generating model?
- Which stylistic features are most discriminative across models?
- How robust are stylistic signatures across different discourse genres?

2.2 Methodology

The methodology follows a four stage pipeline: set up of experiment, corpus generation, feature extraction, and classification with interpretability analysis.

2.2.1 Experimental Setup

For this study, both closed-source and open-source 5 LLMs are chosen. The inclusion of both closed-source and open-source models allows for a comparison of stylistic responses across different level of models. While closed-source models are GPT-4o-mini (OpenAI), Claude Haiku 4.5 (Anthropic) and Gemini 2.5 Flash (Google), open-source models are Qwen2.5-7B-Instruct and LLaMA 3.1-8B-Instruct. This selection provides diversity between model architectures and organizations, while remaining accessible through publicly available APIs without requiring local computational resources. All models were accessed via their respective APIs, with OpenAI API, Anthropic API and Google Generative AI API. Open-source models were accessed through Hugging Face Inference API. To ensure comparability among models, the main generation parameters were set identically across all models. To balance stylistic answer output and respond consistency, temperature value is set as 0.7 in all models. The maximum output length was set to 300 tokens, which is enough for 150 word answers. All models have the same system prompt, instructing them they are a helpful assistant and do not use markdown formatting because of isolating behavioural variation from stylistic differences. “You are a helpful assistant. Do not use markdown formatting, headers, or bullet points in your responses.” Also, for Gemini 2.5 Flash, a model specific parameter is set as thinking budget 0 which disables reasoning phase of model, as reasoning mode was not activated for any model in this study.

2.2.2 Corpus Generation

The corpus consists of 600 texts generated by five LLMs across six genres: explanation, opinion, comparison, narration, argumentation and dialogue. Each model responded same 120 prompts, which makes 120 responses per model with equal number of prompt per genre. These six genres were selected to cover range of communication functions, from analytical discourse to narrative or personal registers, to maximize stylistic variation that models may have for different type of prompts.

Prompt were devoloped with combination of author’s ideas and debating with 3 LLM models, Claude Sonnet 4.6, GPT-5.5 and Gemini. All prompts were reviewed, edited, and finalized by the author regardless of their origin. Each prompt has some fixed set of instructions requiring responses in English, approximately 150 words in length and in paragraph form. The reason of 150 words is decreasing dominant impact of sentence length, which is more behavioral than style, and also because of capabilities of models. Also genre specific instruction were added when it is needed like opinion prompts asked models to express personal view without argumentation, or argumentation has to support one side more, instead of being equal for two side. To further minimize format-driven variation, all models were given the same system prompt disabling markdown formatting, headers, and bullet points. Responses were collected via API calls with temperature set to 0.7 and a maximum output length of 300 tokens across all models.

2.2.3 Feature Extraction

Stylometric features were extracted from each response using two tools: spaCy and StyloMetrix. The final feature set has 14 features, which were selected based on their discriminative power. Features with a zero rate above 80% were excluded, as they provide insufficient variation for discriminative analysis. With spaCy (`en_core_web_sm`), basic stylometric features were computed: average sentence length, punctuation frequency, and part-of-speech ratios for nouns, verbs, adjectives and adverbs. StyloMetrix [1], was used to extract additional features. From available features the following were selected: lemma-based type-token ratio (`ST_TYPE_TOKEN_RATIO_LEMMAS`), active and passive voice ratios (`G_ACTIVE`, `G_PASSIVE`), present and past tense ratios (`G_PRESENT`, `G_PAST`), first-person singular pronoun frequency (`L_I_PRON`), and discourse marker subcategories for contradiction and agreement (`PS_CONTRADICTION`, `PS_AGREEMENT`). Syntactic depth was not included as a standalone feature, as grammatical tense and voice ratios from StyloMetrix partially capture syntactic variation. A correlation analysis confirmed that no feature pair exceeded a correlation coefficient of 0.70, satisfying the independence assumption for Random Forest.

2.2.4 Classification

For the authorship attribution task, a Random Forest classifier was trained to predict which model generated a given text based on the 14 stylometric features. Random Forest was selected because it does not require feature scaling, provides built-in feature importance scores, and handles non-linear relationships between features without strong distributional assumptions.

The model was evaluated using 5-fold stratified cross-validation with `StratifiedKFold`, which preserves class distribution across folds.

Hyperparameters were tuned using grid search. Instead of selecting the configuration with highest cross-validation accuracy, a balanced score was used to penalize overfitting:

$$\text{balanced\_score} = \text{CV accuracy} - 0.5 \times \text{overfit gap} \tag{1}$$

where

$$\text{overfit gap} = \text{training accuracy} - \text{cross-validation accuracy}. \tag{2}$$

The final configuration selected was `max_depth=3`, `min_samples_leaf=5`, `n_estimators=50`.

SHAP values were computed to interpret the classifier's decisions. SHAP values are showing not only which features are most discriminative but also in which direction they push predictions for each model class.

3 Experimental Results

3.1 Classification Results

The Random Forest classifier achieved an overall accuracy of 0.54 and a macro-averaged F1-score of 0.52, significantly above the 20% random baseline for a five-class classification problem. The train accuracy was 0.67, resulting in an overfit gap of 0.13, which is acceptable given the dataset size. Table 1 shows per-class precision, recall and F1-score for each model.

Table 1: Classification results per model

Model	**Precision**	**Recall**	**F1-score**
Claude	0.61	0.86	0.71
Gemini	0.53	0.45	0.49
GPT	0.49	0.37	0.42
LLaMA	0.55	0.71	0.62
Qwen	0.43	0.30	0.35
Macro avg	**0.52**	**0.54**	**0.52**

Claude achieved the highest F1-score (0.71), indicating the most distinctive stylistic profile among all models. LLaMA ranked second (0.62), while Qwen showed the lowest

F1-score (0.35). It shows that its stylistic patterns are the least consistent and hardest to distinguish from other models in Qwen.

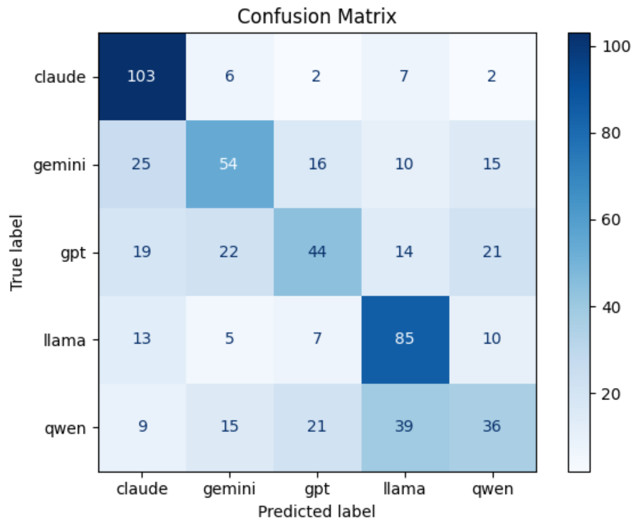


Fig. 1: Confusion Matrix

The confusion matrix revealed that Qwen was most frequently misclassified as LLaMA, and GPT was most frequently confused with Gemini & Qwen, suggesting stylistic proximity between these model pairs.

3.2 Feature Importance and SHAP Analysis

Feature importance scores from the Random Forest model revealed that lexical diversity (0.23), average sentence length (0.19), and punctuation frequency (0.14) are the three most discriminative stylistic features.

Figure 3 shows SHAP summary plots for Claude and Qwen as representative examples of the highest and lowest F1-scoring models respectively. SHAP plots for remaining models are available in the project repository. LLaMA is characterized by low lexical diversity and longer average sentence length. Qwen is distinguished by longer sentence structures and lower active voice usage compared to other models. Claude stands out with its direct writing style, combining short and concise sentences with high active voice usage. Gemini is characterized by rich vocabulary and high punctuation frequency, which likely reflects its tendency toward formatted responses. GPT differentiates itself with high lexical diversity, longer sentences, and heavy noun usage.

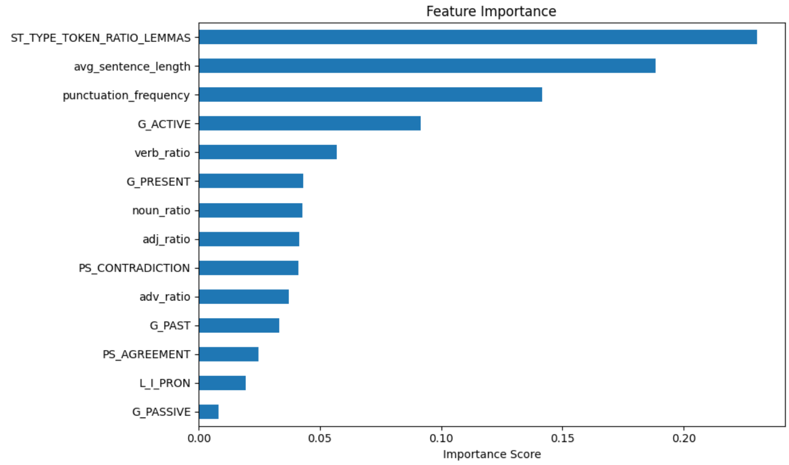
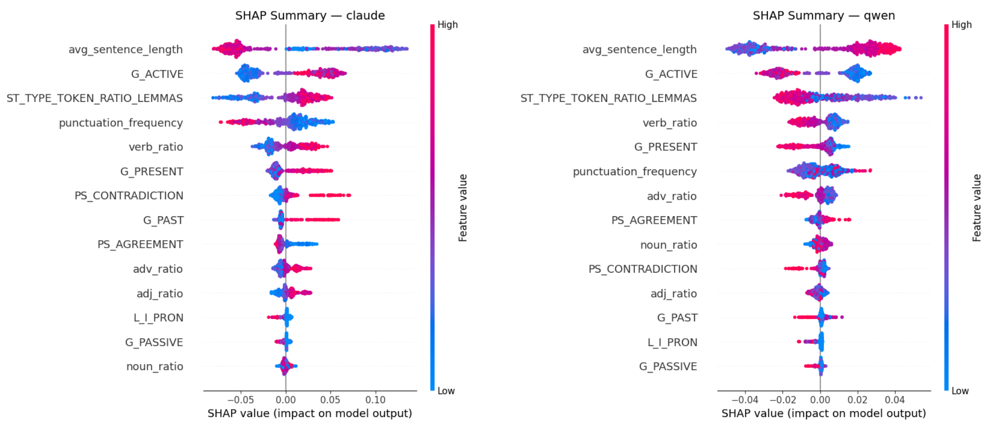


Fig. 2: Feature Importance Scores



(a) Claude

(b) Qwen

Fig. 3: SHAP Summary Plots

3.3 Genre Analysis

To evaluate the robustness of stylistic signatures across different genre, classifier performance was analyzed separately for each genre. Table 2 shows the accuracy per genre alongside the overall accuracy baseline.

Accuracy ranged from 0.44 for explanation to 0.59 for comparison. Explanation genre showed the lowest accuracy, which is expected since technical explanations tend to follow similar structures across models, reducing stylistic variation. Comparison genre showed the highest accuracy, likely because discourse markers such as “whereas” and “on the other hand” create more distinctive patterns per model. Overall, the

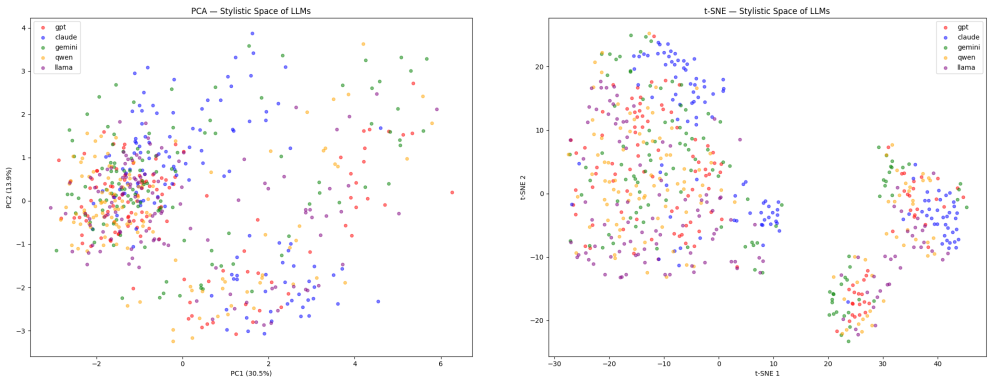
Table 2: Classifier Accuracy by Genre

Genre	Accuracy
Explanation	0.44
Opinion	0.54
Comparison	0.59
Narration	0.52
Argumentation	0.57
Dialogue	0.56
Overall	**0.54**

results suggest that stylistic signatures are partially preserved across genres, but their strength depends on the discourse type.

3.4 Stylistic Space Visualization

To visualize how models are distributed in the stylistic feature space, both PCA and t-SNE dimensionality reduction were applied to the 14-feature matrix.



(a) PCA (b) t-SNE

Fig. 4: Stylistic Space Colored by Model

PCA explained 30.5% of variance in the first component and 13.9% in the second, totaling 44.4%. Both PCA and t-SNE show that models do not form clearly separated clusters in the stylistic space, which is consistent with the classifier accuracy of 0.54. However, when the same t-SNE projection is colored by genre instead of model, a very different picture emerges.

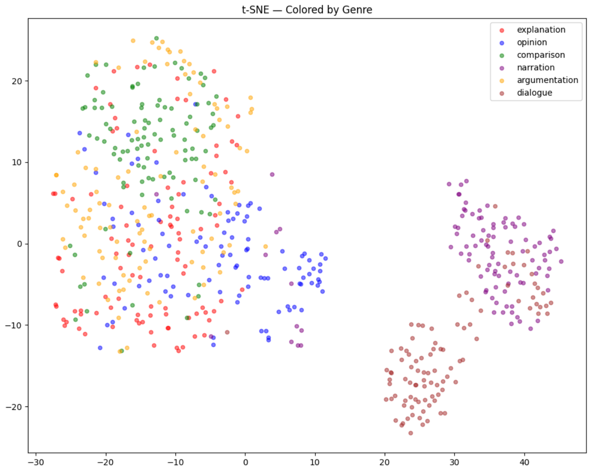


Fig. 5: t-SNE — Stylistic Space Colored by Genre

Dialogue and narration genres form clearly separate clusters from the rest, while explanation, opinion, comparison and argumentation overlap significantly. This suggests that genre has a stronger effect on stylistic space than model identity, and that genre-driven variation may partially mask model-level stylistic differences.

4 Concluding Remarks

This study investigated whether Large Language Models have an identifiable writing style. A Random Forest classifier trained on 14 stylometric features achieved 54% accuracy, which is significantly above the 20% random baseline, showing that LLMs do produce stylistically distinguishable outputs. Among five models, Claude had the most distinctive style with shorter sentences, higher active voice usage and more frequent use of contradiction markers. LLaMA was characterized by low lexical diversity and longer sentences. Gemini showed high punctuation frequency alongside rich vocabulary. GPT and Qwen both produced longer sentences with high lexical diversity, which explains why they were frequently confused with each other. However, t-SNE visualization showed that genre effects are stronger than model effects in the stylistic space. Dialogue and narration genres formed clearly separate clusters regardless of which model produced the text. This suggests that genre-driven variation partially masks model-level stylistic differences. There are some limitations to acknowledge. The dataset size of 600 texts is modest, and the feature set represents only a subset of possible stylometric dimensions. Future work could expand the corpus, apply genre controls in the classification pipeline, and test stylistic robustness under human-mimicry prompting conditions.

AI Usage Disclaimer

Parts of this project were developed with the assistance of Claude Sonnet 4.6 (Anthropic) and GPT-5.5 (OpenAI). The AI was used to support the development of project ideas, the structuring of methodological workflows, the drafting of descriptive texts, and editing the grammar of the report. All content produced with AI assistance has been carefully reviewed, edited, and validated by the author. The author takes full responsibility for the final content and its accuracy, relevance, and academic integrity.

References

[1] I. Okulska, J. Kocoń, and M. Ogrodniczuk. StyloMetrix: A multilingual toolkit for computational stylistic analysis. *SoftwareX*, 24:101545, 2023.

[2] C. Opara. StyloAI: Distinguishing AI-generated content with stylometric analysis. *arXiv preprint arXiv:2405.10129*, 2024.

[3] T. Kumarage, J. Garland, A. Bhattacharjee, K. Trapeznikov, S. Ruston, and H. Liu. Stylometric detection of AI-generated text in Twitter timelines. *arXiv preprint arXiv:2303.03697*, 2023.